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Long Short-Term Memory Recurrent Neural Network for Tidal Level Forecasting

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• **ABSTRACT** Tide is a phenomenon of water level change caused by gravity. Tidal level forecasting is not only a key theoretical topic but also crucial in coastal and ocean engineering applications. The waiting time before a cargo ship enters a port affects the efficiency of cargo transportation, the tidal difference affects the establishment of turbine generators, and an excessive tidal water level reduces vessel safety. With the proliferation of information technology, the application of deep learning models in the analysis and study of hydrological problems has become increasingly common. This study proposed a deep learning model to predict the tidal water level. A forecasting model was developed on the basis of the long short-term memory (LSTM) recurrent neural network for predicting the water levels of 17 harbors in Taiwan. Tidal water level data for 21 years were collected from different observation stations. To objectively evaluate model performance, the developed model was compared with six other forecasting models in terms of the mean absolute percentage error (MAPE) and root mean square error (RMSE) of the forecasting results. The results indicated that the LSTM model had the lowest forecasting error for the tidal water level for up to 30 days. The average MAPE and RMSE values for the developed model were 6.97% and 0.049 m, respectively; thus, the model could effectively reduce the overlapping problems caused by machine learning methods in continuous forecasting.

• **INDEX TERMS** Deep learning, long short-term memory, tidal level forecasting, machine learning.